



## Lab Report: Light Polarization and Malus's Law

Student Name: \_\_\_\_\_

Date: \_\_\_\_\_

### 1. Theory

Malus's Law describes the intensity  $I$  of light transmitted through a polarizer:

$$I = I_0 \cos^2 \theta$$

### 2. Data Collection

Record the relative intensity (or visual brightness) at different angles.

| Angle $\theta$ (degrees) | Relative Intensity ( $I/I_0$ ) | Observed Oscillation Direction |
|--------------------------|--------------------------------|--------------------------------|
| 0°                       | 100%                           |                                |
| 30°                      |                                |                                |
| 45°                      |                                |                                |
| 60°                      |                                |                                |
| 90°                      | 0%                             |                                |

### 3. Analysis Questions

1. At what angle is the light completely blocked (extinction)? Explain why this occurs in terms of wave components.
2. If you place a third polarizer at 45° between two crossed polarizers (90°), does light pass through? Describe what you see in the simulation.